Project Report (Online Learning with Feedback Graphs)

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1 Introduction

The paper studies a generalization of adversarial online learning where the feedback received by the learner is structured by a graph called the feedback graph. In the standard adversarial bandit setup, we have

- K arms
- T rounds
- At each round t, we choose arm $i_t \in [K]$ and pay a cost c_{ti_t}

The goal is to minimize the regret, defined as $R(T) = \sum_{t=1}^{T} c_{ti_t} - \min_{i \in [K]} \left(\sum_{t=1}^{T} c_{ti} \right)$

The feedback graph is a predefined directed graph $G = (V, E)$ for an instance where $V = [K]$, that is the set of vertices corresponds to the set of arms, and E is a set of directed edges. If there is an edge from vertex i to vertex j, then when we choose arm i, we get to observe the cost of arm j. For example,

- In the full information setting, the feedback graph is a complete directed graph with self-loops in all vertices, meaning that when we choose any arm, we get to observe the cost of all arms.
- In the bandit setting, the feedback graph has only self-loops, meaning that when we choose an arm, we only get to observe the cost of that arm.

The paper studies the regret of online learning algorithms in this setting and provides upper and lower bounds on the regret in terms of the structure of the feedback graph. We want to understand how the structure of the feedback graph controls the difficulty of the learning problem.

2 Problem Setting and Definitions

Suppose we are given an adversarial bandit instance with K arms, T rounds, an unknown sequence of cost vectors $c_1, c_2, \ldots, c_T$ where $c_t \in [0,1]^K$ for $t \in [T]$ and a feedback graph $G = (V, E)$ where $V = [K]$. Note that G is fixed and known to the learner.
For $i \in [K]$, define $N^{\text{in}}(i) = \{j \in V \mid (j,i) \in E\}$ as the set of in-neighbors of vertex i and $N^{\text{out}}(i) = \{j \in V \mid (i,j) \in E\}$ as the set of out-neighbors of vertex i. Thus, in each round t, the player pulls arm i_t and observes $\{c_{t,j} \mid j \in N^{\text{out}}(i_t)\}$.
Note that if $(i,i) \in E$, then $i \in N^{\text{in}}(i)$ and $i \in N^{\text{out}}(i)$.

We define the following types of vertices in the feedback graph:

- Observable vertex: $i \in V$ is observable if $N^{\text{in}}(i) \neq \emptyset$, that is there is at least one edge coming into vertex i.
- Strongly observable vertex: $i \in V$ is strongly observable if either i has a self-loop, that is $(i,i) \in E$, or $N^{\text{in}}(i) = V \setminus \{i\}$, that is there is an edge from every other vertex to vertex i.
- Weakly observable vertex: $i \in V$ is weakly observable if it is observable but not strongly observable, that is $N^{\text{in}}(i) \neq \emptyset$ and $(i,i) \notin E$ and there exists $j \in V \setminus \{i\}$ such that $(j,i) \notin E$.

We define the following feedback graphs:

- Observable graph: A feedback graph G is observable if every vertex in G is observable.
- Strongly observable graph: A feedback graph G is strongly observable if every vertex in G is strongly observable.
- Weakly observable graph: A feedback graph G is weakly observable if it is observable but not strongly observable, that is there exists at least one weakly observable vertex in G.

We define the following graph parameters:

- Independence number ($\alpha(G)$): Independent number is the size of the largest independent set in G, that is the largest set of vertices such that there are no edges between any two vertices in the set. $\alpha(G) = \max_{S \subseteq V \mid S \text{ is an IS}} |S|$.
- Weak domination number ($\delta(G)$): Let, $W \subseteq V$ be the set of weakly observable vertices in G. A set $D \subseteq V$ is dominates W if $\forall\, w \in W$, $\exists\, d \in D$ such that $(d,w) \in E$. The weak domination number $\delta(G)$ is the size of the smallest set that dominates W, that is $\delta(G) = \min_{D \subseteq V \mid D \text{ dominates } W} |D|$.

3 Result

Theorem 1. *Given an adversarial bandit instance with feedback graph G and $T \geq |V|^3$ rounds, the expected regret is*

- $\tilde{\Theta}(\alpha(G)^{1/2}T^{1/2})$ *if G is strongly observable*
- $\tilde{\Theta}(\delta(G)^{1/3}T^{2/3})$ *if G is weakly observable*
- $\Theta(T)$ *if G is not observable*

where we ignore poly logarithmic factors in $\tilde{\Theta}$ notation.

4 Upper Bound

The upper bound of $\mathcal{O}(T)$ for unobservable graphs is easy to see since the cost can be at most 1 in each round. For the remaining two cases, we use the EXP3.G algorithm which is a generalization of the EXP3 algorithm for the feedback graph setting.

Algorithm 1 EXP3.G Algorithm

Input: Learning rate $\eta > 0$, exploration parameter $\gamma \in (0, 1]$, feedback graph $G = (V, E)$, exploration set $U \subseteq V$

$u \leftarrow$ uniform distribution over U that is $u_i = \frac{1}{|U|}$ for each $i \in U$ and $u_i = 0$ for each $i \in V \setminus U$
For each arm $i \in V$, set the initial weight $w_{1i} = 1$
for each round $t = 1$ to T **do**
  Compute the probability distribution over arms:

$$p_{ti} \leftarrow (1-\gamma)\frac{w_{ti}}{\sum_{j\in V} w_{tj}} + \gamma u_i \quad \text{for each } i \in V$$

  Sample an arm i_t from the distribution $p_t = (p_{t1}, p_{t2}, \ldots, p_{tK})$
  Observe the costs $\{c_{tj} \mid j \in N^{\text{out}}(i_t)\}$
  for each arm $j \in N^{\text{out}}(i_t)$ **do**
    Compute the estimated cost:

$$\hat{c}_{tj} \leftarrow \frac{c_{tj}}{P_{tj}} \quad \text{where } P_{tj} = \sum_{k\in N^{\text{in}}(j)} p_{tk}$$

    Update the weight of arm j:

$$w_{t+1}(j) \leftarrow w_t(j)e^{-\eta\hat{c}_{t,j}}$$

  end for
end for

<u>Intuition:</u> We update the weights of the arms with the same logic as EXP3 algorithm. Note the $P_{tj} = \sum_{k\in N^{\text{in}}(j)} p_{tk}$ denotes the probability of observing the cost of arm j in round t. Thus, defining $\hat{c}_{tj}$ in this way ensures that the expexted value of $\hat{c}_{tj}$ is equal to c_{tj}, that is $\mathbb{E}[\hat{c}_{tj} \mid H_{t-1}] = c_{tj}$ where H_{t-1} is the history of the algorithm up to round $t-1$.
The idea behind using the exploration set U is that we want to keep those vertices in U that help us to visit the vertices that are hard to explore (Eg: the weakly observable vertices). Even if chooseing vertex from U may be costly, we still want to choose them with some probability to ensure that we get to observe the costs of the hard to explore vertices and thus get a better estimate of the costs of all arms.

Lemma 1. *Let,* $q_{ti} = \dfrac{w_{ti}}{\sum_{j\in V} w_{tj}}$. *Then,* $E[R(T)] \leq \dfrac{\ln K}{\eta} + \eta \displaystyle\sum_{t=1}^{T} E\left[\sum_{i\in V} \frac{q_{ti}}{P_{ti}}\right] + \gamma T$.

Proof. Define $W_t = \sum_{i\in V} w_{ti}$ for each round t. We have,

$$\begin{aligned}
\frac{W_{t+1}}{W_t} &= \frac{\sum_{i\in V} w_{t+1}(i)}{\sum_{i\in V} w_{ti}} \\
&= \sum_{i\in V} \frac{w_{ti}}{\sum_{i\in V} w_{ti}} e^{-\eta\hat{c}_{ti}} \\
&= \sum_{i\in V} q_{ti} e^{-\eta\hat{c}_{ti}} \\
&\leq \sum_{i\in V} q_{ti}(1 - \eta\hat{c}_{ti} + \eta^2\hat{c}_{ti}^2) \qquad [\text{ using } e^{-x} \leq 1 - x + x^2 \text{ for } x \geq 0\,] \\
&= 1 - \eta\sum_{i\in V} q_{ti}\hat{c}_{ti} + \eta^2 \sum_{i\in V} q_{ti}\hat{c}_{ti}^2 \qquad \left[\text{using } \sum_{i\in V} q_{ti} = 1\right]
\end{aligned}$$

$$\implies \ln \frac{W_{t+1}}{W_t} \leq -\eta \sum_{i \in V} q_{ti}\hat{c}_{ti} + \eta^2 \sum_{i \in V} q_{ti}\hat{c}_{ti}^2 \qquad [\text{ using } \ln(1-x) \leq -x \,]$$

$$\implies \ln \frac{W_{T+1}}{W_1} = \sum_{t=1}^{T} \ln \frac{W_{t+1}}{W_t} \leq -\eta \sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti} + \eta^2 \sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti}^2 \qquad [\text{ summing over } t = 1, \ldots, T \,] \qquad (1)$$

For any arm $i^* \in [K]$, $W_{T+1} \geq w_{T+1}(i^*) = e^{-\eta \sum_{t=1}^{T} \hat{c}_{ti^*}}$. Also, $W_1 = \sum_{i \in V} w_{1i} = K$ since $w_{1i} = 1$ for each $i \in [K]$. Thus,

$$\ln \frac{W_{T+1}}{W_1} \geq -\eta \sum_{t=1}^{T} \hat{c}_{ti^*} - \ln K$$

$$\implies -\eta \sum_{t=1}^{T} \hat{c}_{ti^*} - \ln K \leq -\eta \sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti} + \eta^2 \sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti}^2 \qquad [\text{ using } (1) \,]$$

$$\implies \sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti} - \sum_{t=1}^{T} \hat{c}_{ti^*} \leq \frac{\ln K}{\eta} + \eta \sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti}^2 \qquad (2)$$

We know, $E[\hat{c}_{ti}] = \sum_{H_{t-1}} P[H_{t-1}] \cdot E[\hat{c}_{ti} \mid H_{t-1}] = c_{ti} \left(\sum_{H_{t-1}} P[H_{t-1}] \right) = c_{ti}$.

Now, $E[q_{ti}\hat{c}_{ti}] = \sum_{H_{t-1}} P[H_{t-1}] \cdot E[q_{ti}\hat{c}_{ti} \mid H_{t-1}] = \sum_{H_{t-1}} P[H_{t-1}] \cdot E[q_{ti}c_{ti} \mid H_{t-1}] = E[q_{ti}c_{ti}]$.

Also, $E[q_{ti}\hat{c}_{ti}^2] = \sum_{H_{t-1}} P[H_{t-1}] \cdot E[q_{ti}\hat{c}_{ti}^2 \mid H_{t-1}] = \sum_{H_{t-1}} P[H_{t-1}] \cdot E\left[q_{ti}\frac{c_{ti}^2}{P_{ti}} \mid H_{t-1}\right] \leq \sum_{H_{t-1}} P[H_{t-1}] \cdot E\left[\frac{q_{ti}}{P_{ti}} \mid H_{t-1}\right] = E\left[\frac{q_{ti}}{P_{ti}}\right]$

Taking expectation on both sides of (2), we get,

$$E\left[\sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti}\right] - E\left[\sum_{t=1}^{T} \hat{c}_{ti^*}\right] \leq \frac{\ln K}{\eta} + \eta E\left[\sum_{t=1}^{T}\sum_{i \in V} q_{ti}\hat{c}_{ti}^2\right]$$

$$\implies E\left[\sum_{t=1}^{T}\sum_{i \in V} q_{ti}c_{ti}\right] - \sum_{t=1}^{T} c_{ti^*} \leq \frac{\ln K}{\eta} + \eta E\left[\sum_{t=1}^{T}\sum_{i \in V} \frac{q_{ti}}{P_{ti}}\right] \qquad (3)$$

Considering actual probabilities of picking the arms,

$$\sum_{i \in V} p_{ti}c_{ti} = (1-\gamma)\sum_{i \in V} q_{ti}c_{ti} + \gamma \sum_{i \in V} u_i c_{ti} \leq (1-\gamma)\sum_{i \in V} q_{ti}c_{ti} + \gamma \sum_{i \in V} u_i \leq \sum_{i \in V} q_{ti}c_{ti} + \gamma \sum_{i \in V} u_i = \sum_{i \in V} q_{ti}c_{ti} + \gamma$$

Applying this in (3), we get,

$$\implies E\left[\sum_{t=1}^{T}\left(\sum_{i \in V} p_{ti}c_{ti} - \gamma\right)\right] - \sum_{t=1}^{T} c_{ti^*} \leq \frac{\ln K}{\eta} + \eta E\left[\sum_{t=1}^{T}\sum_{i \in V} \frac{q_{ti}}{P_{ti}}\right]$$

$$\implies E\left[\sum_{t=1}^{T}\sum_{i \in V} p_{ti}c_{ti}\right] - \sum_{t=1}^{T} c_{ti^*} \leq \frac{\ln K}{\eta} + \eta \sum_{t=1}^{T} E\left[\sum_{i \in V} \frac{q_{ti}}{P_{ti}}\right] + \gamma T \qquad \text{for all } i^* \in [K]$$

$$\implies E[R(T)] = E\left[\sum_{t=1}^{T}\sum_{i \in V} p_{ti}c_{ti}\right] - \min_{i^* \in [K]} \sum_{t=1}^{T} c_{ti^*} \leq \frac{\ln K}{\eta} + \eta \sum_{t=1}^{T} E\left[\sum_{i \in V} \frac{q_{ti}}{P_{ti}}\right] + \gamma T$$

□

4.1 Strongly observable graphs

We choose the exploration set $U = V$.

Lemma 2. *Let, $G = (V, E)$ be a directed graph with $|V| = K$, in which each node $i \in V$ is assigned a positive weight w_i. Assume that $\sum_{i \in V} w_i \leq 1$ and $w_i \geq \epsilon$ for each $i \in V$ for some constant $0 < \epsilon < \frac{1}{2}$. Then, $\sum_{i \in V} \frac{w_i}{w_i + \sum_{k \in N^{in}(i)} w_k} \leq 4\alpha(G) \ln\left(\frac{4K}{\alpha(G)\epsilon}\right)$.*

Lemma 3. *For strongly observable graphs and $U = V$, $\sum_{i \in V} \frac{q_{ti}}{P_{ti}} = \mathcal{O}(\alpha(\mathcal{G}) \log \mathcal{K})$.*

Proof. Recall that $p_{ti} = (1-\gamma)q_{ti} + \gamma u_i$, and since $U = V$, we have $u_i = \frac{1}{K}$ for every $i \in V$. Hence $p_{ti} \geq \frac{\gamma}{K}$ for every $i \in V$. Also,

$$P_{ti} = \sum_{k \in N^{\text{in}}(i)} p_{tk}.$$

Now apply Lemma 2 with the weights $w_i = p_{ti}$ and $\epsilon = \dfrac{\gamma}{K}$. This gives

$$\sum_{i \in V} \frac{p_{ti}}{P_{ti}} = \sum_{i \in V} \frac{p_{ti}}{p_{ti} + \sum_{k \in N^{\text{in}}(i)} p_{tk}} \leq 4\alpha(G) \ln\left(\frac{4K}{\alpha(G)\epsilon}\right) = 4\alpha(G) \ln\left(\frac{4K^2}{\alpha(G)\gamma}\right).$$

Since $q_{ti} = \dfrac{p_{ti} - \gamma/K}{1-\gamma} \leq \dfrac{p_{ti}}{1-\gamma}$ and we take $\gamma \leq \frac{1}{2}$, we get $q_{ti} \leq 2p_{ti}$. Therefore

$$\sum_{i \in V} \frac{q_{ti}}{P_{ti}} \leq 2 \sum_{i \in V} \frac{p_{ti}}{P_{ti}} \leq 8\alpha(G) \ln\left(\frac{4K^2}{\alpha(G)\gamma}\right).$$

Thus,

$$\sum_{i \in V} \frac{q_{ti}}{P_{ti}} = \mathcal{O}(\alpha(G) \log K),$$

up to logarithmic factors. □

Applying this to lemma 1, we get,

$$E[R(T)] \leq \frac{\ln K}{\eta} + \eta T \cdot \mathcal{O}(\alpha(G) \log K) + \gamma T$$

Taking $\gamma = \eta = \sqrt{\dfrac{1}{\alpha(G) \cdot T}}$ gives us $E[R(T)] = \tilde{\mathcal{O}}(\alpha(G)^{1/2} T^{1/2})$.

4.2 Weakly observable graphs

Let, D be the smallest set that dominates the set of weakly observable vertices W. We choose the exploration set U to be D. Thus, we have $|U| = \delta(G)$. Thus,

$$P_{ti} = \sum_{k \in N^{\text{in}}(i)} p_{tk} \geq p_{td} \geq \frac{\gamma}{\delta(G)} \quad \text{for each } i \in W \text{ and some } d \in D$$

Thus, for weakly observable vertices i, $\dfrac{q_{ti}}{P_{ti}} \leq q_{ti} \cdot \dfrac{\delta(G)}{\gamma}$.

For a strongly observable vertex i, if it has a self loop, then $P_{ti} \geq p_{ti} \geq (1-\gamma)q_{ti} \geq \dfrac{q_{ti}}{2}$ for $\gamma \leq \dfrac{1}{2} \implies \dfrac{q_{ti}}{P_{ti}} \leq 2$

If a strongly observable vertex i does not have a self loop, then it has an incoming edge from every other vertex. So, there is a dominating vertex $d \in D$ such that $(d, i) \in E$. Thus, $P_{ti} \geq p_{td} \geq \dfrac{\gamma}{\delta(G)} \implies \dfrac{q_{ti}}{P_{ti}} \leq q_{ti} \cdot \dfrac{\delta(G)}{\gamma}$.

Thus, for every vertex $i \in V$, $\dfrac{q_{ti}}{P_{ti}} \leq \max\left(2, q_{ti}\dfrac{\delta(G)}{\gamma}\right) \leq 2 + q_{ti}\dfrac{\delta(G)}{\gamma}$.

Applying this to lemma 1, we get,

$$\begin{aligned}
E[R(T)] &\leq \frac{\ln K}{\eta} + \eta \sum_{t=1}^{T} E\left[\sum_{i \in V} 2 + q_{ti}\frac{\delta(G)}{\gamma}\right] + \gamma T \\
&= \frac{\ln K}{\eta} + \eta \sum_{t=1}^{T} E\left[2K + \frac{\delta(G)}{\gamma}\right] + \gamma T \qquad \left[\text{since } \sum_{i \in V} q_{ti} = 1\right] \\
&= \frac{\ln K}{\eta} + \eta T\left(2K + \frac{\delta(G)}{\gamma}\right) + \gamma T
\end{aligned}$$

Taking $\gamma = \left(\dfrac{\delta(G) \ln K}{T}\right)^{1/3}$ and $\eta = \left(\dfrac{\ln K}{T\sqrt{\delta(G)}}\right)^{2/3}$ gives us $E[R(T)] = \tilde{\mathcal{O}}(\delta(G)^{1/3} T^{2/3})$.

5 Lower Bound

5.1 Strongly observable graphs

The lower bound for strongly observable graphs is a known result. In particular, the paper of Alon et al. shows that for every strongly observable graph G,

$$R(G,T) = \Omega(\sqrt{\alpha(G)T}).$$

Therefore,

$$R(G,T) = \tilde{\Omega}(\alpha(G)^{1/2}T^{1/2}).$$

This matches the upper bound proved above up to logarithmic factors.

5.2 Weakly observable graphs

Lemma 4. *Let $G = (V,E)$ be a directed graph over $|V| = n$ vertices, and let $W \subseteq V$ be a set of vertices whose minimal dominating set is of size k. Then, W contains an independent set U of size at least $\frac{k}{50 \ln n}$, with the property that any vertex of G dominates at most $\ln n$ vertices of U.*

Let W be the set of weakly observable vertices in G. By Lemma 4 , there exists an independent set $U \subseteq W$ such that

$$|U| = m = \Omega\left(\frac{\delta(G)}{\ln K}\right),$$

and every vertex of G dominates at most $O(\ln K)$ vertices of U.
Fix $\epsilon = m^{1/3}(32T \ln K)^{-1/3}$. Choose $\chi \in U$ uniformly at random. For each round t and arm i, define the random costs by

$$c_{t,i} \sim \begin{cases} \mathrm{Ber}\left(\frac{1}{2} - \epsilon\right), & i = \chi, \\ \mathrm{Ber}\left(\frac{1}{2}\right), & i \in U,\ i \neq \chi, \\ 1, & i \notin U. \end{cases}$$

Let M_i be the number of times arm $i \in U$ is played, and let N_i be the number of times the player plays an arm that reveals arm i. Also let N be the number of times the player plays an arm outside U.
If $N > \epsilon T$, then the regret is already at least $\Omega(\epsilon T)$, because every arm outside U has loss 1 while every arm in U has expected loss at most $\frac{1}{2}$. So we may assume that $N \leq \epsilon T$.

Since every arm outside U dominates at most $O(\ln K)$ vertices of U, we get

$$\sum_{i \in U} N_i \leq N \ln K \leq \epsilon T \ln K.$$

Now fix $i \in U$. Let Q_i be the distribution of the learner's observations when $\chi = i$, and let Q_0 be the distribution when every arm in U has mean $\frac{1}{2}$.
For a fixed round t, the conditional distributions under Q_i and Q_0 differ only if the learner plays an arm that reveals i. In that case, the KL divergence between the observed Bernoulli distributions is at most $4\epsilon^2$. Hence, by the chain rule,

$$D_{\mathrm{KL}}(Q_0, Q_i) \leq 4\epsilon^2\, E_{Q_0}[N_i].$$

By Pinsker's inequality,

$$D_{\mathrm{TV}}(Q_0, Q_i) \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(Q_0, Q_i)} \leq \epsilon \sqrt{2E_{Q_0}[N_i]}.$$

Averaging over $i \in U$ and using concavity of the square root,

$$\frac{1}{m} \sum_{i \in U} D_{\mathrm{TV}}(Q_0, Q_i) \leq \sqrt{\frac{2\epsilon^2}{m} E_{Q_0}\left[\sum_{i \in U} N_i\right]} \leq \sqrt{\frac{2\epsilon^3 T \ln K}{m}}.$$

By our choice of ϵ, the right-hand side is at most $\frac{1}{4}$. Therefore,

$$\frac{1}{m} \sum_{i \in U} D_{\mathrm{TV}}(Q_0, Q_i) \leq \frac{1}{4}.$$

Now under Q_i, the expected regret is at least $\epsilon\big(T - E_{Q_i}[M_i]\big)$ since every round on which the player does not play $\chi = i$ incurs at least ϵ extra expected loss.
Also,

$$E_{Q_i}[M_i] - E_{Q_0}[M_i] \leq T \cdot D_{\mathrm{TV}}(Q_0, Q_i).$$

Averaging over $i \in U$ gives

$$\frac{1}{m}\sum_{i\in U} E_{Q_i}[M_i] \le \frac{1}{m}\sum_{i\in U} E_{Q_0}[M_i] + T\cdot\frac{1}{m}\sum_{i\in U} D_{\mathrm{TV}}(Q_0, Q_i).$$

Since $\sum_{i\in U} M_i \le T$, we have

$$\frac{1}{m}\sum_{i\in U} E_{Q_0}[M_i] \le \frac{T}{m}.$$

Using the bound on total variation,

$$\frac{1}{m}\sum_{i\in U} E_{Q_i}[M_i] \le \frac{T}{m} + \frac{T}{4} \le \frac{3T}{4}, \qquad [\text{ because } m \ge 2\ ]$$

$$\text{Therefore,} \quad E[R(T)] \ge \epsilon\left(T - \frac{3T}{4}\right) = \frac{\epsilon T}{4}.$$

Substituting the value of ϵ, $E[R(T)] = \Omega\big(m^{1/3}T^{2/3}(\ln K)^{-1/3}\big)$. Since $m = \Omega\Big(\frac{\delta(G)}{\ln K}\Big)$, we obtain $E[R(T)] = \tilde{\Omega}\big(\delta(G)^{1/3}T^{2/3}\big)$.

5.3 Non observable graphs

Assume that G is not observable. Then there exists a vertex, say 1, such that $N^{\text{in}}(1) = \emptyset$. Hence the learner never observes the cost of arm 1.

Now choose $\chi \in \{0, 1\}$ uniformly at random and define $c_{t,1} = \chi$ for all t, and $c_{t,i} = \frac{1}{2}$ for all t and all $i \ne 1$. Since arm 1 is never observed, the learner's sequence of played arms is independent of χ. Let M be the number of times the learner plays arm 1.

If $\chi = 0$, then the best arm is 1 and the regret is $\sum_{t=1}^{T} c_{t,i_t} - 0 = \frac{1}{2}(T - M)$.

If $\chi = 1$, then the best arm is any arm other than 1, and the regret is $\sum_{t=1}^{T} c_{t,i_t} - \frac{T}{2} = \frac{1}{2}M$.

Taking expectation over χ,

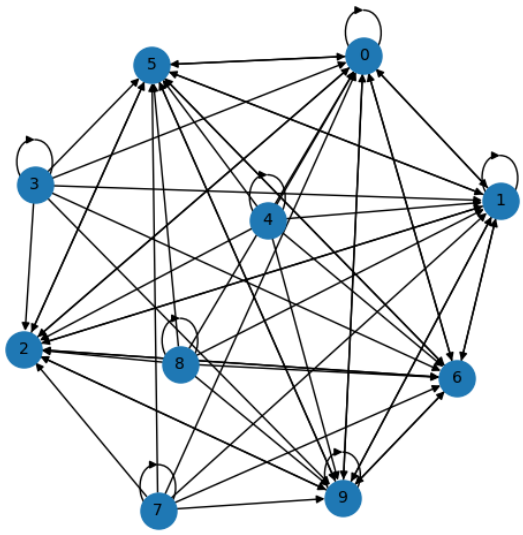
$$E[R(T)] = \frac{1}{2}E\left[\frac{1}{2}(T - M)\right] + \frac{1}{2}E\left[\frac{1}{2}M\right] = \frac{T}{4}.$$

Therefore, for every player algorithm there exists a loss sequence such that $E[R(T)] \ge \frac{T}{4}$, which proves $R(G, T) = \Theta(T)$.

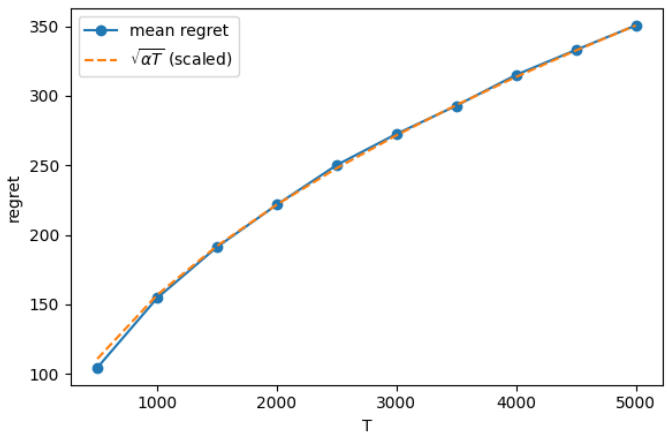
6 Experimental Results

6.1 Strongly Observable graph

We select a random strongly observed graph with $K = 10$ arms and run the EXP3.G algorithm for $T = [500, 1000, \ldots, 5000]$ rounds taking this graph as the feedback graph of the problem. For each T, we run the algorithm for 200 independent trials and plot the average regret across these trials. We also plot the function $\alpha(G)^{1/2}T^{1/2}$ for comparison.



(a) A random strongly observable graph with $K = 10$ arms



(b) Regret vs T for the strongly observable graph with $K = 10$ arms repeated for 200 independent trials

Figure 1: Strongly observable graph and corresponding regret plot

6.2 Weakly Observable graph

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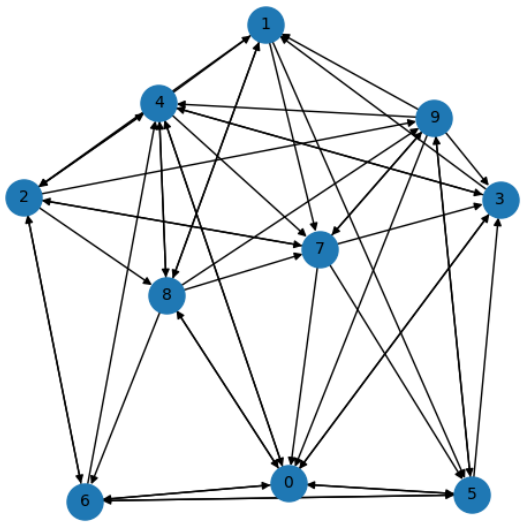
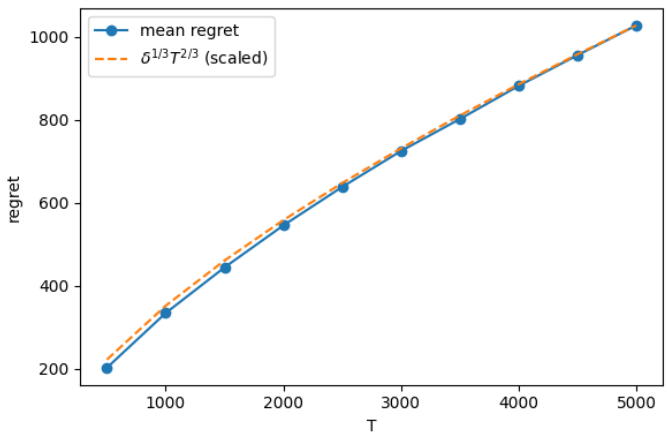


(a) A random weakly observable graph with $K = 10$ arms

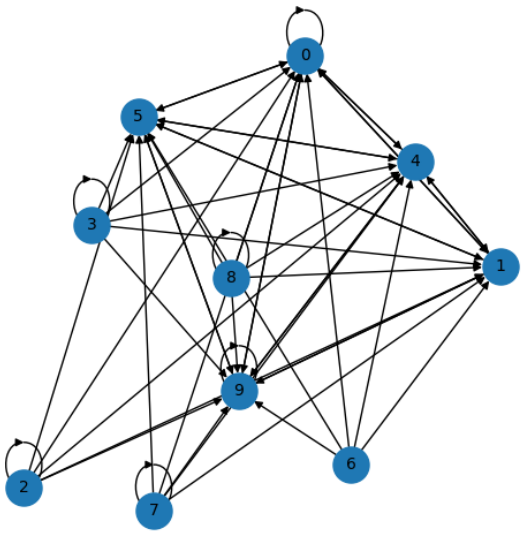


(b) Regret vs T for the weakly observable graph with $K = 10$ arms repeated for 200 independent trials

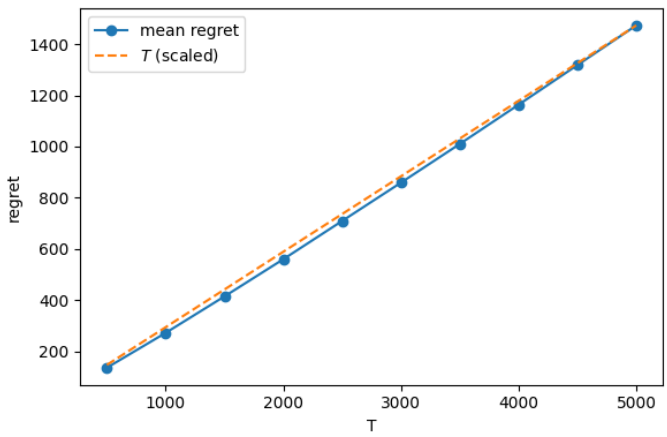
Figure 2: Weakly observable graph and corresponding regret plot

6.3 Non Observable graph

We pick a random non-observable graph with $K = 10$ arms and run the EXP3.G algorithm for $T = [500, 1000, \ldots, 5000]$ rounds taking this graph as the feedback graph of the problem. For each T, we run the algorithm for 200 independent trials and plot the average regret across these trials. We also plot the function T for comparison.



(a) A random non-observable graph with $K = 10$ arms



(b) Regret vs T for the non-observable graph with $K = 10$ arms repeated for 200 independent trials

Figure 3: Non-observable graph and corresponding regret plot

Remark: The code for these experiments is available here

7 References

[1] Noga Alon, Nicolò Cesa-Bianchi, Ofer Dekel, and Tomer Koren. Online Learning with Feedback Graphs: Beyond Bandits. arXiv preprint arXiv:1502.07617, 2015.